

# VITHURUSHAN MEENESWARAN

**AI Tech Lead** · Founder, GenZ Innovations (Pvt) Ltd

1 Coopers Road, Potters Bar, London EN6 1JG · +44 7774 946384 · [vithurushan@gmail.com](mailto:vithurushan@gmail.com)  
[linkedin.com/in/vithurushan](https://linkedin.com/in/vithurushan) · [github.com/Siva130497](https://github.com/Siva130497)

## PROFILE

AI Tech Lead with seven years in software, now building agentic systems full time. Founder of GenZ Innovations, where I architect and build Velaris — a kernel-mediated AI operating system spanning a Rust kernel, a Bun LLM router, Python agent and evaluation services, and a Next.js shell, reaching users across nine channels.

My focus is the hard half of autonomy: least-privilege tool scoping, human approval bound to exactly what was approved, untrusted content fenced so it cannot reach instruction position, and deterministic checks running ahead of any model judgement. Before founding, I was tech lead for MERN delivery at Prodigit — including Skillety, an AI recruitment platform now live across India — while working as a senior engineer for DigiFront in Norway.

## CORE SKILLS

|                                   |                                                                                                                   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>AI orchestration</b>           | LangGraph, LangChain, DeepAgents, MCP, Vercel AI SDK, multi-agent systems, tool calling, human-in-the-loop        |
| <b>Models &amp; inference</b>     | OpenAI, Anthropic, Google, OpenRouter, fal.ai, Hugging Face, ONNX Runtime, streaming / SSE                        |
| <b>AI safety &amp; evaluation</b> | Prompt-injection defence, least-privilege tool scopes, approval binding, eval pipelines, LLM-as-judge, guardrails |
| <b>LLM observability</b>          | Langfuse, OpenTelemetry, Sentry, cost attribution, distributed tracing                                            |
| <b>Languages</b>                  | TypeScript, JavaScript, Python, Rust, PHP, Dart, Kotlin, C++, SQL                                                 |
| <b>Backend &amp; platform</b>     | Node.js, Bun, NestJS, Hono, Express, FastAPI, Laravel, Livewire, gRPC, BullMQ                                     |
| <b>Front end</b>                  | React, Next.js, Astro, Tailwind CSS, GSAP, Figma                                                                  |
| <b>Data</b>                       | PostgreSQL, Supabase, Prisma, MongoDB, MySQL, Redis                                                               |
| <b>Mobile &amp; systems</b>       | Flutter, Android (Kotlin), Linux, macOS, Windows, networking, Arduino                                             |
| <b>Tooling</b>                    | Docker, Turborepo, pnpm, Git, Postman, Vercel, Fly.io                                                             |

## EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Founder &amp; AI Tech Lead</b> — GenZ Innovations (Pvt) Ltd, London, UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025 – Present  |
| <ul style="list-style-type: none"><li>Architect and build Velaris, an AI operating system across four runtimes — a Rust kernel, a Bun LLM router, Python agent and evaluation services, and a Next.js OS shell</li><li>Designed the agent supervision layer: plans compiled before execution, with risk and tool scopes derived from a registry rather than inferred by a model, and ceremony scaled to the consequence of the action</li><li>Own the safety model — least-privilege scoping, approval receipts cryptographically bound to the approved plan, and untrusted-content fencing against prompt injection</li><li>Route and observe LLM spend and quality across OpenAI, Anthropic, Google and fal.ai using Langfuse and OpenTelemetry</li></ul> |                 |
| <b>MERN Stack Tech Lead</b> — Prodigit, India (Remote)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jun 2023 – 2025 |
| <i>Held concurrently with the DigiFront role below — both remote.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |
| <ul style="list-style-type: none"><li>Led the team building Skillety, an AI-driven recruitment platform matching uploaded CVs to 2,400+ live openings from 140+ companies</li><li>Set React and Node.js architecture and code standards for a distributed team</li><li>Ran review, release and production support end to end</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |
| <b>Senior Software Engineer</b> — DigiFront AS, Norway (Remote)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mar 2023 – 2025 |
| <i>Held concurrently with the Prodigit role above — both remote.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |
| <ul style="list-style-type: none"><li>Owned end-to-end delivery across React front ends and Node.js services</li><li>Worked asynchronously across time zones, from specification through review to release</li><li>Maintained and extended production codebases alongside greenfield work</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |

**Software Engineer** — F11 Software Labs, Colombo, Sri Lanka

Nov 2020 – Mar 2023

- Shipped full-stack features across booking platforms, e-commerce storefronts and their admin dashboards
- Designed and integrated REST APIs against MySQL and MongoDB
- Delivered cross-platform mobile features in Flutter
- Reviewed code and mentored junior and associate engineers

**Associate Software Engineer** — F11 Software Labs, Colombo, Sri Lanka

Sep 2019 – Oct 2020

- Implemented UI components and pages from design handoff
- Wrote back-end endpoints and database queries under senior review
- Fixed defects and improved performance on live client applications

**Intern Software Engineer** — F11 Software Labs, Colombo, Sri Lanka

Mar 2019 – Aug 2019

- Built static and dynamic pages with HTML, CSS and JavaScript
- Learned version control, code review and agile delivery in practice

**SELECTED PROJECTS**

---

**Velaris** · AI operating system · 2026 – ongoing

Kernel-mediated platform where every model call, agent run and channel message passes one audited spine that enforces permission, cost and audit before anything leaves the network.

*Rust · TypeScript · Python · LangGraph · Next.js · Supabase · gRPC*

**Skilllety** · [skilllety.com](https://skilllety.com) · 2023 – 2025

AI recruitment platform: candidates upload a CV and an AI recruiter matches them to live roles and generates an ATS-ready CV; employers post openings and search candidates.

*React · Node.js · Express · MongoDB*

**Greencore Order Planner** · 2026

Turns a weekly stock-wastage PDF into next week's order list entirely in the browser, matched against 358 product lines extracted from 366 planogram placements.

*React · pdf.js · Vite*

**GREYSKULL** · 2025

Installable PWA for forecourt shift management — station diary, essential stock, staffing, issue logging and a live weather banner.

*React · Express · MongoDB · Zod · PWA*

**The Parking Deals** · [theparkingdeals.co.uk](https://theparkingdeals.co.uk) · 2023

A complete redesign and rebuild of the earlier Airwing Parking platform. Compares meet-and-greet, park-and-ride and long-stay parking across Heathrow, Gatwick and Luton on a live quote, with promo codes and a full back-office admin behind it.

*React · Node.js · Express · MongoDB*

**CCTV Motion Capture System** · Personal project

An Arduino security camera watching a 180° arc with PIR sensors; on detection a servo swings the camera base toward the movement and LED status lights show which sensor fired.

*Python · C++ · Arduino · PIR sensors*

**Wi-Fi File Transfer** · Personal project

Turns an Android phone into an FTP server so its storage mounts on Android, Windows, macOS and iOS over the local network — or remotely, with a forwarded port.

*Kotlin · Java · FTP · Android*

**EDUCATION**

---

**MSc Artificial Intelligence and Robotics** — University of Hertfordshire, UK

2025 – 2026

**BSc (Hons) Software Engineering** — Solent University, Southampton, UK

2022 – 2023

**Higher National Diploma in Information Technology** — Advance Institute of Technology, Jaffna

2019 – 2021

**Diploma in Hardware, Mobile & CCTV** — Edhat International**G.C.E. Advanced Level — Biology stream** — Victoria College, Sri Lanka**G.C.E. Ordinary Level — 9A passes** — Victoria College, Sri Lanka**ADDITIONAL**

---

**Languages** English, Tamil

**Teaching** 4+ years teaching the MERN stack (freelance)

**Interests** Agentic system design, AI safety and red-teaming; provincial-level chess champion